

Root phone (POCO X5 5G)

Privacy Analysis of Augmented Reality Glasses

1 Unlocking the Bootloader

Note that Xiaomi provides official USB drivers for Windows, which are required for both fastboot and Mi Flash Tool to properly detect the phone. Other systems do not have native drivers for Xiaomi devices, and needs to be configured but can still have compatibility issues, therefore a Windows computer is what was used (and is preferred) for the unlocking the bootloader.

Steps to unlock bootloader

On the phone:

- Make sure to have Xiaomi account that is linked to the phone number of the sim card and that the number has mobile data. The steps to creating this account can be found in the section below.
- Become a developer by going to '**About phone**' and tapping '**MIUI version**' 7 times until the following message appears: You are now a developer.
- Go to '**Additional settings**', '**Developer options**',
- Enable USB debugging and OEM unlocking.
- Turn off wifi and enable cellular data from the sim card.
- Go back to '**Additional settings**', '**Developer options**' and go to '**Mi unlock status**'.
- Agree to the permissions and press '**Add account and device**'. It should a successful message that you were able to add the device at the bottom of the screen.

Now go onto the computer:

- Install Mi Unlock Tools on the computer, need windows (unofficial Mi Unlock tool).
- Extract the file you downloaded.
- Double tap MiUsbDriver from the extracted folder to extract.
- Double tap MiUsbDriver to launch application.
- You should see that the opened window says that phone is not connected.

Now back onto the phone:

- Power off the phone.
- With the Xiaomi Poco X5 powered off, press and hold the '**Power**' + '**Volume Down**' buttons until you see the Xiaomi Poco X5 Fastboot mode screen.
- Connect the Xiaomi Poco X5 phone to your PC / computer with an original USB cable.

Now onto the computer:

- In your opened window, it should now start to install drivers.
- After the installation is completed, go back to the folder for the Mi Unlock Tools.
- Double tap miflash_unlock to extract.

- Double tap `miflash_unlock` to launch application.
- Log in to mi unlock (better to use phone number for log in info so they don't ask for verification with phone number, hence won't need to reboot your phone then power it off and put into fastboot mode again)
- App should show: Phone connected. Then press '**Unlock**'.
- There could be a mandatory waiting time, which is 7 days, before you can unlock the bootloader. Try to make sure the phone does not run out of batteries, nor log out of the Xiaomi account as that can reset the timer.
- You should see the window saying '**Unlocked successfully**'. Then press '**Reboot phone**'.
- After setting up the phone, go to '**Mi Unlock Status**' in settings (remember you will need to make yourself a developer again), it should show that the device is unlocked.

2 Rooting the phone

Something to be cautious is that although TWRP is a very common approach for rooting a phone, it did not work for me. Usually, the rooting process of a device with an unlocked bootloader would involve installing a custom recovery such as TWRP and then flashing Magisk directly as a ZIP file. However, the phone I am using uses dynamic partitions, and TWRP had trouble working with these partitions, and couldn't mount (or access) some parts of the system like `/data` or `/storage`. This caused errors like:

```
Unable to mount '/data'
```

Therefore, I chose another method by patching and flashing the '**boot.img**' with Magisk directly, therefore bypassing the need for TWRP.

Note that Xiaomi provides official USB drivers for Windows, which are required for both '**fastboot**' and '**Mi Flash Tool**' to properly detect the phone. Other systems do not have native drivers for Xiaomi devices, and needs to be configured but can still have compatibility issues. Therefore a Windows computer is what was used (and is preferred) for the rooting the phone.

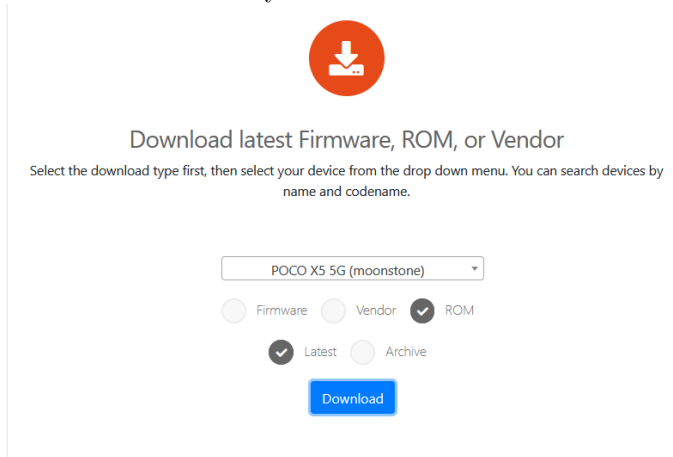
I also used an SD card, so if you don't have an SD card holder for the phone, this method is not viable. If there is no SD card holder on the computer, you can use another one to transfer files to the SD card.

Steps to root phone

On the computer that has an SD card holder:

- Plug the SD card into your computer.
- Install the latest stable Magisk APK from the link:
<https://github.com/topjohnwu/Magisk/releases>
- Transfer the Magisk apk onto the SD card. You can do this by directly dragging and dropping the file onto the mounted SD card in file explorer.
- Download the fastboot ROM (it is also in case of any issues with the phone where any partition or files get corrupted, so the phone becomes dead and you are unable to reboot the phone). This allows you to restore the phone. It can be found with the following link:
<https://xmfirmwareupdater.com/>.

- In the window choose your device model and the rest like so



- Press '**Download**' on the window
- On the new page, find a '**device**' (it should be the device you selected but you need to choose a region) that is closest to your location, mine was '**EEA**'. Make sure that the '**branch**' is '**stable**' and the '**type**' is '**Fastboot**'. Then press download.
- Scroll down in the new window and choose one of the displayed '**Full ROM Mirror #**' to download. If one takes more than half an hour, switch to another one as they are simply different services for the same download.
- After download, extract the .tgz file, then extract the .tar file inside it. For Windows, you may need to install '**7-zip**' to extract the files.
- In the extracted folder, go into '**images**' and transfer '**boot.img**' onto the SD card.

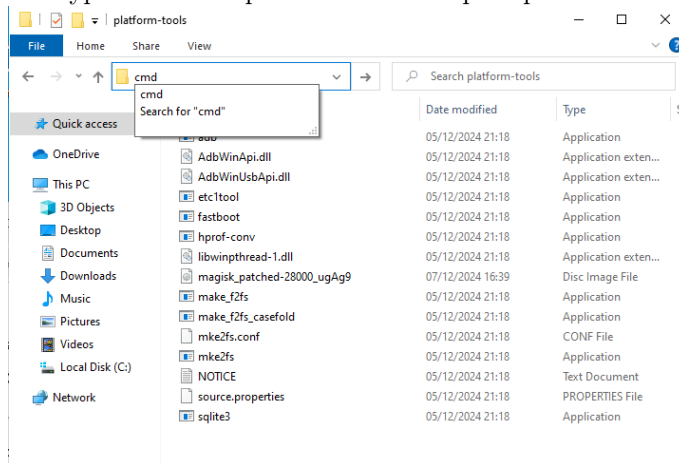
On the phone:

- Make sure that you are a developer in settings. To do this, tap the build number 7 times in '**About phone**'.
- Then make sure to have USB debugging enabled in '**Additional settings**', then '**Developer options**'.
- Plug the SD card into the phone.
- Go to '**File Manager**' and go to where the '**Magisk APK**' is located (it should be in storage/sdcard).
- Tap on the APK to install it, and if prompted enable '**Install Unknown Apps**' in the phone settings. This should start the download of the app on your phone.
- Open the app, and under the word '**Magisk**', you should see something along the lines: '**Magisk is not installed**'.
- Tap '**Install**' to the right side of '**Magisk**'. Then '**Select and patch a file**'.
- Go to the location of the '**boot.img**' on the phone (it should be in storage/sdcard), where Magisk will then process and create a patched version of the file. This file will be saved as '**magisk_patched.img**' in the '**Downloads**' folder of the phone.
- Connect the phone to the windows computer and make sure that it is in 'File Transfer' mode.

On the Windows computer:

- Install '**android platform tools**' (adb and fastboot). Note that if you unlocked the bootloader using the steps from the section '**Unlocking the Bootloader**', this should already be installed and you can directly go to the next step. If not, install it from the link: <https://developer.android.com/tools/releases/platform-tools>. Then extract the downloaded zip file.

- Go to the extracted '**platform tools**' folder.
- On the top left corner of '**File Explorer**' where the path to the folder is shown, tap in there and type '**cmd**' to open the command prompt in in that directory. Like so



- Check if the connection to the phone is successful by writing the following in the command prompt:

```
adb devices
```

You should see your phone. If not, recheck the USB connectivity.

- Copy the patched magisk file from your phone to your PC using the command:

```
adb pull /sdcard/Download/magisk_patched.img
```

Note that the file will be saved in the same folder as the current directory in the command prompt.

- The '**magisk_patched.img**' and your '**fastboot**' executable from the platform-tools directory should be in the same directory.

Now on the phone:

- Power off the phone.
- Go into fastboot mode by pressing and holding the '**Power**' + '**Volume Down**' button on the side of the phone until the Fastboot logo appears.

Back onto the Windows computer:

- Check if your phone is detected by writing the following in the command prompt:

```
fastboot devices
```

You should see your phone. If not, recheck the USB connectivity.

- Flash the patched image by using the command:

```
fastboot flash boot magisk_patched.img
```

- After flashing is successful, reboot the phone into the system:

```
fastboot reboot
```

This could take a few minutes.

- Once the phone is booted and set up, open the Magisk app.
- It should now say that Magisk is installed with whichever Magisk version you downloaded.
- Another way to verify is to download a '**Root Check**' app from the '**Play Store**'. If you do install the app, when the app prompts for root permissions, you will need to grant it.

3 Create Xiaomi Account

The problem for me when creating an account was the phone number.

Initially, I was not able to create an account using a phone number because I kept getting the error that the number was not valid even though it was, so I used the mail to create an account and added the phone number afterwards.

Then when I did create an account using mail, I was not able to find the place where I can add a phone number on the phone nor online. So I used the following link add my phone number:

<https://account.xiaomi.com/pass/serviceLogin?checkSafePhone=true>

which I got from the thread:

https://www.reddit.com/r/LineageOS/comments/1d6y974/how_do_i_add_phone_number_to_xiaomi_account/

Note that when I tried to do this initially on my computer, it kept saying that the captcha I did to prove I am not a bot is incorrect, so I tried the link on my phone where it worked.

Steps to creating an account

1. On the phone go to 'Xiaomi Account' in settings.
2. Tap on create account and follow the on screen instructions. (Note that you will need a valid mail account for this).
3. If you do use a mail for creating an account, you will then need to link the phone number of your sim card to that Xiaomi account. To do this, go to the following link and link the phone number to your account: <https://account.xiaomi.com/pass/serviceLogin?checkSafePhone=true>. If it does not work, try the link on another device.

4 Reflashing the phone

If you end up corrupting any of the partitions and are not able to boot into the system you can restore your device. Reflashing the firmware will not relock the bootloader unless the script used was not '**flash_all_lock.bat**' script (or similar commands to relock the bootloader).

- Download the official Fastboot ROM for the phone model if not done yet through this link: <https://xmfirmwareupdater.com/>
- Put phone into fastboot mode by pressing the '**Power**' + '**Volume Down**' button.
- On the computer, copy the image you plan to flash into the directory of '**fastboot.exe**' and go to that directory.
- Run the flash script '**flash_all.bat**' to reflash the entire firmware by double clicking on the file if on Windows, otherwise use the command:

```
./flash_all.sh
```

This process could take a couple minutes.

- Reboot into the system after flashing:

```
fastboot reboot
```

If you only need to flash a single partition, you can flash each individually. Note that some of the partitions could be in '**super.img**' which requires one to extract the file to get the partitions images inside.

Examples of the commands look like (for the boot, system, vendor, and data partitions):

```
fastboot flash boot boot.img
fastboot flash system system.img
fastboot flash vendor vendor.img
fastboot flash userdata userdata.img
```

5 Rooting via TWRP

Steps to rooting the phone via TWRP

1. Connect the phone to the computer via USB
2. Four things need to be downloaded before we can root the phone.
 - (a) Download the TWRP image for the phone model. One can find it by directly typing the following into the search bar:

`twrp for <phone model>`

The official website should be starting with '[twrp.me/...](https://twrp.me/)'.

Click the region closer to you in the download links and then install the latest img file.

- (b) The sdk tools is also needed. This should already be downloaded during unlocking of the bootloader, but if not, it can be downloaded using the following link:

<https://developer.android.com/tools/releases/platform-tools>

- (c) Install '**Disable_Dm-Verity_ForceEncrypt**'. Open to the link:

<https://zackptg5.com/android.php#disverfe>

There should be a '**Dm-Verity, Forced Encryption, and Disc Quota Disabler**' on the webpage. Open '**Dm-Verity & ForceEncrypt Disabler**' and click download.

- (d) Install Magisk zip from github, using the link:

<https://github.com/topjohnwu/Magisk/releases>

Rename the file extension to .zip if necessary.

3. Open a terminal in the sdk platform tools directory.
4. Power off the phone then reboot into fastboot mode by pressing on the '**Power**' + '**Volume Down**' button.
5. Check if your phone is detected by writing the following in the command prompt:

```
fastboot devices
```

6. Flash TWRP using Fastboot with the command onto the boot partition:

```
fastboot flash boot twrp.img
fastboot reboot recovery
```

Do not disconnect the phone during this process.

7. Go to '**Wipe**', then '**Format Data**' then type '**yes**' to confirm.
8. Now on computer, go to the internal storage of you phone and copy paste '**Disable_Dm-Verity_ForceEncrypt.zip**' in there.

If that does not work, try transferring the files using the command:

```
adb push Disable_Dm-Verity_ForceEncrypt.zip /sdcard
adb push Magisk.zip /sdcard
```

9. Back onto the phone, go back to '**Install**' and select the Disable_Dm-Verity_ForceEncrypt zip file and swipe to flash.
10. Reboot the phone by first going back and tap on '**Reboot**', '**Recovery**' to refresh the partition table.
11. Go to '**Install**' and select the Magisk.zip and swipe to flash.
12. Reboot the phone into the system by tapping on '**Reboot**', '**System**'.

Problems Encountered

Phone does not boot

- Phone does not boot into TWRP

This could either be because TWRP wasn't installed correctly or your device is booting from the wrong slot.

If its the installation, then flash TWRP again and reboot:

```
fastboot flash boot twrp.img
fastboot reboot recovery
```

If its the slot, first check which is the current slot and switch to the other one:

```
fastboot getvar current-slot
fastboot set_active a # Or b
```

- Phone does not boot into System

This could be because of slot mismatch but it is most likely because some of the necessary partitions or files are corrupted.

If its the slot, first check which is the current slot and switch to the other one:

```
fastboot getvar current-slot
fastboot set_active a # Or b
```

Otherwise, the method to solve this is to reflash the corrupted partitions or the entire ROM to restore the system. The steps to this can be found in section 4 **Reflashing the phone**.

Installation problems due to commands

- When installing TWRP:

When initially trying to install TWRP using the commands:

```
fastboot boot C:\Users\michelle\Downloads\twrp-3.7.0_12-1-moonstone.img
```

I get the following error:

```
Sending 'boot.img' (131072 KB)          OKAY [2.995s]
Booting                               FAILED (remote: 'unknown command')
fastboot: error: Command failed
```

This means that my device or the fastboot tool does not support the '**fastboot boot**' command. Verify that the TWRP image is compatible with your device, and use the command:

```
fastboot flash boot C:\Users\michelle\Downloads\twrp-3.7.0_12-1-moonstone.img
```

- When installing TWRP:

Another command tried was:

```
fastboot flash recovery C:\Users\michelle\Downloads\twrp-3.7.0_12-1-moonstone.img
```

I got the error:

```
Warning: skip copying recovery image avb footer (recovery partition size: 0, recovery ima
Sending 'recovery' (131072 KB)          OKAY [3.016s]
Writing 'recovery'                     FAILED (remote: '(recovery_b) No such partition'
fastboot: error: Command failed
```

This means that the device does not have a 'traditional' recovery partition. Many newer Xiaomi devices with A/B partition, including the POCO X5 5G, schemes lack a dedicated recovery partition, which means TWRP needs to be installed differently by using the following command instead:

```
fastboot flash boot C:\Users\michelle\Downloads\twrp-3.7.0_12-1-moonstone.img
```

Permission issues

- When wiping Data Format:
When wiping the data, the log showed the error:

```
Unable to mount storage
Full SELinux support is present.
Unable to locate /data.
```

This could be either due to partition corruption or encryption.

Methods to resolve it include trying to reflash the partition if corrupted, or changing the file system if encrypted, using the command:

```
fastboot format:ext4 userdata
```

where I got this error:

```
C:\Users\michelle\Downloads\platform-tools-latest-windows\platform-tools>fastboot format:
FAILED (remote: 'unknown command')
fastboot: error: Command failed
```

This shows that that my device does not support the fastboot format command, so I tried using another command:

```
fastboot erase userdata
```

where I got this error:

```
C:\Users\michelle\Downloads\platform-tools-latest-windows\platform-tools>fastboot erase u
Erasing 'userdata'                               FAILED (remote: 'Check device console.
fastboot: error: Command failed
```

This means that there is likely a restriction imposed by the bootloader or firmware, preventing direct modification of the userdata partition.

- When installing Magisk:
The log showed an error:

```
Unable to repack boot image
```

This could be due to the device using a dynamic partition scheme, which can complicate Magisk installation via TWRP, or due to encryption or restricted write access to the boot partition.

Methods to resolve this could be to patch and flash the boot image on another device where one can install Magisk (this is the method I went for in the end that did not require the use of TWRP), or to try and format data which erase all internal storage.

- When installing TWRP:
The log showed:

```
Unable to mount storage error
```

The reason is due to encryption issues on the phone's storage, meaning that TWRP may not be able decrypt your internal storage because it doesn't have the proper decryption keys for the device.

Methods I tried included reinstalling the '**Disable_Dm-Verity_ForceEncrypt**' file. As well as rewiping format data to erase all files in internal storage. I also tried to change the file system.

- When installing Magisk:
The log showed:

```
Unable to patch ramdisk
Unmounting partitions
Updater process ended with ERROR: 1 Error installing zip file '/sdcard/Magisk-v28.0.z
```

This is due to the data partition being encrypted or TWRP does not have the correct permissions to access certain partitions.

Methods I tried was to rewipe the Format Data, and go to '**Mount**' to check if the '**Data**' partition is available. It was not for me which suggests a deeper problem where TWRP is unable to detect the internal storage and data partition layout of the device. This issue is common on devices with dynamic partitions or encryption schemes unsupported by the installed TWRP version.

To resolve this, I first checked the TWRP version I installed to make sure it was compatible with my phone model. That was not the problem so I tried to reformat the '**Data**' partition using the command:

```
fastboot erase userdata
```

where I got this error:

```
C:\Users\michelle\Downloads\platform-tools-latest-windows\platform-tools>fastboot erase u
Erasing 'userdata'                                     FAILED (remote: 'Check device console.
fastboot: error: Command failed
```

This means that there is likely a restriction imposed by the bootloader or firmware, preventing direct modification of the userdata partition.